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COURSE NAME: CLOUD COMPUTING AND BIG DATA ANALYTICS

ASSESSMENT TOOL-5

APPLICATION BASED QUESTIONS

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.

Ans) Cloud Auto-Scaling and Load Balancing

A company that experiences sudden traffic spikes during seasonal sales can use cloud auto-scaling and load balancing to maintain smooth application performance. Auto-scaling automatically increases the number of servers when traffic rises and reduces them when demand decreases. This ensures that sufficient computing resources are always available without wasting money.

Load balancing distributes incoming user requests evenly across multiple servers. Instead of sending all requests to one server, it shares the workload, preventing server overload and improving response time. If one server fails, the load balancer redirects traffic to healthy servers, reducing downtime.

Monitoring tools such as Amazon CloudWatch, Azure Monitor, or Google Cloud Monitoring continuously track CPU usage, memory utilization, network traffic, and response times. When predefined thresholds are reached, these tools automatically trigger scaling actions and send alerts to administrators.

A real-world example is Amazon during festive sales such as Prime Day or Flipkart during Big Billion Days. Millions of users visit simultaneously, and cloud auto-scaling with load balancing ensures the website remains fast and available, providing customers with a smooth shopping experience.

2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.

Ans) Cloud Storage, Security, and Disaster Recovery

Organizations can securely store large volumes of backup data using cloud storage services such as Amazon S3, Azure Blob Storage, or Google Cloud Storage. These services provide high durability by storing multiple copies of data across different servers and geographical locations. Even if one storage device fails, data remains accessible from another replica.

Encryption protects sensitive backup data by converting it into unreadable form. Data can be encrypted both while being transmitted (in transit) and while stored (at rest). Access control mechanisms such as Identity and Access Management (IAM), Multi-Factor Authentication (MFA), and Role-Based Access Control (RBAC) ensure that only authorized users can access backup files.

Compared to traditional storage systems, cloud storage eliminates the need for purchasing hardware, maintenance, cooling, and infrastructure upgrades. Organizations pay only for the storage they use, making cloud storage highly cost-effective.

A disaster recovery example is a company recovering its databases after a flood damages its local data center. Since backups are stored securely in the cloud, systems can be restored quickly, ensuring business continuity with minimal downtime.

3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

Ans) CDN and Global Application Deployment

When a mobile application is deployed globally, Content Delivery Networks (CDNs) improve performance by delivering content from servers located closest to users. Instead of accessing data from a single central server, users receive content from nearby edge servers, significantly reducing latency.

Edge servers store cached copies of frequently accessed content such as images, videos, application files, and web pages. This minimizes the distance data travels, resulting in faster loading times and better user experience.

Major cloud providers such as AWS, Microsoft Azure, and Google Cloud maintain global data centers connected through CDNs, enabling high availability and fault tolerance. If one region experiences problems, traffic can automatically be redirected to another region.

A practical example is Netflix or YouTube, where video content is delivered through CDN edge servers. Users experience smooth video streaming with minimal buffering regardless of their geographical location. Similarly, online multiplayer games use CDNs to reduce lag and improve gameplay.

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data

sovereignty requirements are met? Give an example involving sensitive financial transactions.

Ans) Cloud Compliance and Security in Financial Institutions

Financial institutions require strict security and regulatory compliance while using cloud services. Cloud providers support compliance standards such as ISO 27001, PCI DSS, GDPR, HIPAA, and SOC certifications. These standards ensure that cloud environments meet industry security requirements.

Logging and monitoring services continuously record user activities, login attempts, configuration changes, and system events. Security teams analyze these logs to detect suspicious behavior and perform audits. Encryption protects sensitive financial information during storage and transmission, ensuring customer data remains confidential.

Data sovereignty requirements can be satisfied by selecting cloud regions within a specific country so that sensitive financial information never leaves national boundaries. Cloud providers allow organizations to choose the geographic location where their data is stored.

For example, during online banking transactions, customer account information and payment details are encrypted. Every transaction is logged for auditing, while monitoring tools detect fraud or unauthorized access attempts, ensuring secure financial operations.

5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Ans) Containers, Kubernetes, and Microservices

Containers package an application together with its libraries, dependencies, and configuration files. Unlike Virtual Machines (VMs), containers share the host operating system kernel, making them lightweight and much faster to start. Containers consume less CPU, memory, and storage compared to VMs.

Virtual Machines include a complete guest operating system for each instance, increasing resource consumption and startup time. Containers therefore provide better performance and higher resource utilization.

Container orchestration platforms such as Kubernetes automate deployment, scaling, monitoring, networking, and recovery of containerized applications. Kubernetes automatically distributes containers across servers, restarts failed containers, and scales applications based on workload.

Containers are highly portable because they can run consistently across different cloud providers and operating systems without modification. This allows organizations to avoid vendor lock-in and easily migrate applications between AWS, Azure, Google Cloud, or on-premises environments.

A good example is a microservices-based e-commerce application, where separate containers handle user authentication, product catalog, payment processing, order management, and notifications. Kubernetes manages all these services efficiently, ensuring high availability, automatic scaling, and reliable performance.